

## AKTOBE, Kazakhstan

WMO: 352290

Lat: 50.2578N

Lon: 57.2236E

Elev: 219

StdP: 98.72

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -29.0      | -26.2 | -32.1                         | 0.2 | -28.8 | -29.3 | 0.3 | -26.3 | 13.4                  | -6.2 | 12.2 | -8.9 | 1.7                      | 150  | 0.608 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.5                   | 35.6            | 18.5 | 33.9 | 18.3 | 31.9 | 17.8 | 20.4                | 30.6 | 19.6 | 29.5 | 18.9 | 28.6 | 4.5                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.3                            | 12.7 | 22.5 | 16.3 | 11.9 | 22.5 | 15.5 | 11.3 | 22.3 | 59.5          | 30.5 | 56.8 | 29.3 | 54.4 | 28.5 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 12.1              | 10.4 | 9.2 | DB                         | -33.1 | 38.1               | 3.9 | 1.7                                                | -35.9 | 39.4       | -38.2 | 40.4       | -40.4 | 41.4       | -43.3 | 42.6 |  |
|                   |      |     | WB                         | -33.2 | 22.1               | 3.9 | 0.8                                                | -36.0 | 22.7       | -38.3 | 23.2       | -40.5 | 23.7       | -43.3 | 24.3 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 5.6    | -13.3 | -12.0 | -4.7  | 7.6   | 15.3  | 20.7  | 22.7  | 21.3  | 14.4  | 6.3   | -2.3  | -9.7  | (6)  |
|                                                                        | DBStd       | 14.09  | 7.00  | 7.05  | 6.14  | 6.04  | 4.50  | 4.34  | 3.61  | 4.34  | 4.68  | 4.82  | 6.01  | 6.94  | (7)  |
|                                                                        | HDD10.0     | 3049   | 722   | 617   | 457   | 115   | 9     | 1     | 0     | 0     | 13    | 134   | 369   | 612   | (8)  |
|                                                                        | HDD18.3     | 5030   | 981   | 850   | 715   | 323   | 116   | 24    | 6     | 21    | 133   | 372   | 619   | 870   | (9)  |
|                                                                        | CDD10.0     | 1446   | 0     | 0     | 1     | 43    | 172   | 321   | 392   | 352   | 145   | 19    | 0     | 0     | (10) |
|                                                                        | CDD18.3     | 386    | 0     | 0     | 0     | 1     | 21    | 95    | 140   | 114   | 15    | 0     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 5356   | 0     | 0     | 0     | 42    | 371   | 1323  | 1794  | 1537  | 282   | 7     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 2428   | 0     | 0     | 0     | 6     | 113   | 616   | 843   | 750   | 99    | 1     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 3.5    | 3.4   | 3.7   | 4.0   | 4.0   | 3.8   | 3.5   | 3.2   | 3.1   | 3.1   | 3.3   | 3.5   | 3.6   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 305    | 22    | 20    | 21    | 27    | 29    | 35    | 30    | 25    | 20    | 29    | 26    | 26    | (15) |
|                                                                        | PrecMax     | 406    | 55    | 67    | 47    | 60    | 80    | 99    | 109   | 75    | 75    | 82    | 62    | 66    | (16) |
|                                                                        | PrecMin     | 193    | 3     | 0     | 0     | 0     | 3     | 0     | 0     | 3     | 0     | 0     | 3     | 0     | (17) |
|                                                                        | PrecStd     | 60     | 13    | 15    | 14    | 17    | 21    | 23    | 23    | 20    | 15    | 17    | 13    | 16    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 1.5   | 2.0   | 14.8  | 27.5  | 31.9  | 36.8  | 37.2  | 38.0  | 34.2  | 24.0  | 12.0  | 3.1   | (19) |
|                                                                        |             | MCWB   | 0.9   | 1.0   | 7.6   | 13.8  | 16.4  | 18.6  | 19.1  | 18.3  | 16.7  | 12.4  | 8.6   | 2.0   | (20) |
|                                                                        | 2%          | DB     | 0.1   | 0.7   | 7.8   | 24.1  | 29.8  | 34.9  | 35.1  | 35.2  | 29.1  | 20.8  | 9.1   | 1.3   | (21) |
|                                                                        |             | MCWB   | -0.3  | 0.1   | 4.8   | 12.8  | 16.5  | 18.0  | 19.0  | 18.5  | 14.9  | 10.9  | 6.6   | 0.7   | (22) |
|                                                                        | 5%          | DB     | -1.2  | -0.4  | 4.6   | 21.0  | 27.8  | 32.8  | 33.1  | 33.2  | 26.7  | 17.9  | 7.1   | 0.6   | (23) |
|                                                                        |             | MCWB   | -1.7  | -1.0  | 2.6   | 11.5  | 15.9  | 17.8  | 18.4  | 18.1  | 14.2  | 9.8   | 5.4   | 0.1   | (24) |
|                                                                        | 10%         | DB     | -3.0  | -1.8  | 2.7   | 18.1  | 25.0  | 30.2  | 31.2  | 31.0  | 23.8  | 14.8  | 5.0   | -0.8  | (25) |
|                                                                        |             | MCWB   | -3.4  | -2.4  | 1.4   | 10.4  | 14.7  | 17.1  | 18.2  | 17.4  | 13.5  | 8.6   | 3.5   | -1.4  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 0.9   | 1.1   | 8.2   | 15.2  | 19.1  | 20.8  | 21.5  | 21.2  | 18.2  | 14.1  | 9.9   | 2.1   | (27) |
|                                                                        |             | MCDB   | 1.2   | 1.5   | 13.2  | 23.9  | 27.5  | 31.3  | 31.7  | 33.0  | 29.8  | 21.2  | 11.1  | 2.5   | (28) |
|                                                                        | 2%          | WB     | -0.3  | 0.1   | 4.9   | 13.7  | 17.7  | 19.6  | 20.6  | 20.0  | 16.1  | 12.0  | 7.2   | 0.9   | (29) |
|                                                                        |             | MCDB   | 0.0   | 0.5   | 7.4   | 22.0  | 26.7  | 30.1  | 30.3  | 30.8  | 25.4  | 18.7  | 8.4   | 1.2   | (30) |
|                                                                        | 5%          | WB     | -1.8  | -1.1  | 2.7   | 12.2  | 16.6  | 18.8  | 19.9  | 19.0  | 15.0  | 10.6  | 5.3   | -0.1  | (31) |
|                                                                        |             | MCDB   | -1.4  | -0.6  | 4.2   | 19.6  | 25.2  | 28.9  | 29.3  | 29.5  | 24.2  | 16.0  | 6.7   | 0.5   | (32) |
|                                                                        | 10%         | WB     | -3.5  | -2.3  | 1.4   | 10.7  | 15.5  | 18.0  | 19.1  | 18.2  | 14.0  | 9.2   | 3.5   | -1.5  | (33) |
|                                                                        |             | MCDB   | -3.0  | -1.7  | 2.4   | 17.3  | 23.6  | 27.7  | 28.4  | 28.4  | 22.2  | 14.1  | 4.7   | -0.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.3   | 9.3   | 9.1   | 11.9  | 13.7  | 14.0  | 13.5  | 14.5  | 13.9  | 11.4  | 7.2   | 7.6   | (35) |
|                                                                        |             | MCDBR  | 6.1   | 6.5   | 9.6   | 16.9  | 17.8  | 17.2  | 16.4  | 17.6  | 18.5  | 17.3  | 9.5   | 6.3   | (36) |
|                                                                        |             | MCWBR  | 6.1   | 6.1   | 6.9   | 8.2   | 7.7   | 6.5   | 6.2   | 7.1   | 8.7   | 9.8   | 7.2   | 5.8   | (37) |
|                                                                        | 5% WB       | MCDBR  | 6.3   | 6.2   | 8.7   | 15.3  | 16.1  | 15.0  | 14.1  | 15.9  | 15.9  | 15.0  | 8.2   | 5.8   | (38) |
|                                                                        |             | MCWBR  | 6.4   | 6.0   | 6.6   | 7.9   | 7.5   | 6.1   | 5.6   | 6.6   | 8.0   | 9.0   | 6.8   | 5.6   | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.298 | 0.326 | 0.384 | 0.427 | 0.412 | 0.410 | 0.412 | 0.426 | 0.373 | 0.355 | 0.334 | 0.302 | (40) |
|                                                                        | taud        |        | 2.088 | 2.024 | 1.971 | 2.074 | 2.212 | 2.283 | 2.288 | 2.220 | 2.324 | 2.296 | 2.195 | 2.097 | (41) |
|                                                                        | Ebn at Noon |        | 700   | 778   | 796   | 810   | 845   | 849   | 841   | 805   | 814   | 753   | 656   | 628   | (42) |
|                                                                        | Edh at Noon |        | 83    | 115   | 146   | 148   | 135   | 127   | 125   | 127   | 103   | 88    | 75    | 71    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.39  | 2.58  | 3.81  | 4.75  | 6.01  | 6.66  | 6.41  | 5.49  | 3.99  | 2.21  | 1.03  | 0.90  | (44) |
|                                                                        | RadStd      |        | 0.16  | 0.19  | 0.48  | 0.47  | 0.55  | 0.49  | 0.32  | 0.39  | 0.31  | 0.32  | 0.14  | 0.18  | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | +0.88   | -0.39 | N/A   | N/A         | N/A     | +105    | +84     |
| (47) | Regional (3 neighbors) | N/A   | N/A     | N/A    | +0.88   | -0.39 | N/A   | N/A         | N/A     | +105    | +84     |

Nomenclature: See separate page